

ITA0506-COMPUTER VISION LAB EXPERIMENT

DATE :15-07-2026

EXPNO:3

AIM: Perform basic Image Handling and processing operations on the image

• Read an image in python and Convert an Image to show outline using Canny function.

Algorithm:

- import library
Start the program and import the OpenCV module using `import cv2`.
- Read the input image
Read the image from the given file path using `cv2.imread()`.
- Check image loading
Verify that the image is loaded (not None); if loading fails, print an error message and stop the program.
- Convert to grayscale
Convert the color image to a grayscale image using `cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)`.
- Filter noise and detect edges
Apply Gaussian blur on the grayscale image using `cv2.GaussianBlur()` and then perform Canny edge detection using `cv2.Canny()` to obtain the outline.
- Display images and exit
Display the original and edge-detected images using `cv2.imshow()`, wait for a key press with `cv2.waitKey(0)`, then close all windows using `cv2.destroyAllWindows()`.

Code:

```
*ita0506 lab 3.py - C:/Users/surya/OneDrive/Desktop/surya/ita0506 lab 3.py (3.13.5)*
File Edit Format Run Options Window Help
import cv2

img = cv2.imread(r"C:\Users\surya\OneDrive\Pictures\Screenshots 1\Screenshot 2026-07-23 112236.png")

if img is None:
    print("Error: Image not found or unable to open.")
    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

blur = cv2.GaussianBlur(gray, (5, 5), 0)
edges = cv2.Canny(blur, 100, 200)
cv2.imshow("Original Image", img)
cv2.imshow("Canny Outline", edges)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT :

```
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